

CURRICULUM VITAE

Xavier PIC, PhD
POSTDOCTORAL RESEARCHER

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Sophia Antipolis, France

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BIOGRAPHY

I am a postdoctoral researcher at EURECOM, Sophia Antipolis, France, since November 2024, under the supervision of Raja Appuswamy, in the Data Science Team. My activities principally cover the field of image coding with a focus on AI-based methods as well as methods adapted to DNA data storage, which was my PhD's topic. I design constrained coders (entropy, fixed-length, arithmetic coders) that are adapted to the specificities of DNA data storage, and use them to modify or implement novel image coders adapted to DNA data storage. These image coders have to both be performant (rate distortion optimization) and robust to the noise generated by the biochemical operations of DNA data storage. As a member of the JPEG Group, I have contributed to the JPEG DNA Ad-hoc research group, for the definition of an image coding standard respecting the specificities of DNA data storage. I am also following the work in the JPEG AI Ad-Hoc group.

EDUCATION

FALL 2024	PhD in Image Compression, I3S Laboratory (Côte d'Azur University and CNRS - UMR 7271), Sophia Antipolis, France Joint CNRS/Imperial College of London funding
FALL 2019	Engineering Degree in Computer Science, ENSEIRB (Bordeaux Institute of Technology), Bordeaux, France
SUMMER 2016	Preparatory scientific courses for engineering schools College: Lycée Masséna, Nice, France Classes préparatoires de mathématiques supérieures
SUMMER 2015	English formation in Cambridge, United Kingdom
JUNE 2014	French High-School Degree with mathematics specilization General High school: Lycée général Jean Moulin, Draguignan, France Baccalauréat scientifique spécialité mathématiques With Distinction

POSITIONS HELD IN THE PAST

SUMMER 2026 - PRESENT	Postdoc	I3S Laboratory, Sophia Antipolis, France Postdoctoral researcher position on data compression methods for DNA data storage
FALL 2024 - SPRING 2026	Postdoc	EURECOM, Sophia Antipolis, France Postdoctoral researcher position on AI-based image compression methods
SPRING 2021 - FALL 2024	PhD	I3S Laboratory, Sophia Antipolis, France Coding algorithms for long-term storage of digital images on synthetic DNA molecules.
SPRING - SUMMER 2019	Internship	Allab, Università degli Studi di Firenze, Italy Research internship in Deep Learning applied to the understanding of Handwritten Document Analysis
SUMMER 2018	Internship	SURYS, Bussy-Saint Georges, FRANCE Research internship on Deep Learning and Image Processing methods to classify digital images of passports and find weak authenticating signals

OTHER PAST PROJECTS

FALL 2021 - PRESENT	Involvement in the JPEG DNA Ad-Hoc group ISO/IEC JTC 1/SC 29/WG 1
FALL 2018	Study of Deep Learning solutions for Super Resolution for the Bordeaux Bioinformatics Center (CBIB)
FALL 2017 - SPRING 2018	Creation of performance and quality tests on CAFFE models for the Bordeaux Informatics Research Laboratory (LaBRI)
WINTER 2017-2018	Creation of a compiler for pseudocode C in LLVM code

RESEARCH

Key words Images, Deep Learning, compression, multimedia, constrained coding, wavelets, compressive autoencoders, learning-base image compression, DNA data storage, analysis of DNA encoded data, oligo formatting

Main Activities My research activity focuses on the study of performant compression methods adapted to image coding. I have a specific interest in learning-based approaches of the problem of image compression. I also have designed compression methods that are adapted to the field of DNA data storage. They require coders that are both adapted to the coding constraints imposed by the DNA biochemical operations and robust to the noise occurring in these lossy biochemical operations. The main topics of this work are:

- **Image compression**
 - Rate distortion optimization
 - Noise robustness
 - DCT/DWT-based image coding methods (JPEG, JPEG200), Learning-based image coding methods (autoencoders, implicit neural representation)
- **Constrained coding**
 - Entropic, arithmetic, fixed length coders on incomplete quaternary codes
 - Close-to-balanced coders on incomplete quaternary codes
- **DNA data storage** - (*contributor to the JPEG DNA standardization group*)
 - Design coders that take into account the constraints imposed by the DNA biochemical processes of synthesis, storage and sequencing
 - Design of robust to noise quaternary codes (representing DNA sequences) using the A, T, C, G alphabet
 - Formatting the quaternary codes into oligos (short strands of DNA) of fixed lengths
 - Design codecs robust to errors occurring at the oligo level (oligo shuffling, oligo loss)

Publications The results obtained through my work have been published in one journal (Pattern Recognition Letters), and in several international congresses and workshops (see my publication list below). I am currently finalizing the redaction of an article to be submitted in IEEE Transactions on Image Processing. I also have participated in the redaction of standardization document output by the JPEG DNA Ad-hoc group.

Software The software that I have been producing for experiments during my thesis aim at improving the state of the art in DNA data storage. Some of those software have been provided to the JPEG DNA ad-hoc group.

- Goldman Transcoder – *JPEG contribution (JPEG DNA compression and coding Anchor 1)*
- JPEG DNA Benchmark Codec – *JPEG contribution (JPEG DNA compression and coding Anchor 2)* : DNA Image codec provided to the JPEG DNA Ad-hoc group as an Anchor to compare the results of the proponents in the JPEG DNA Call for Proposals
- DNAAutoenc : Compressive Autoencoder encoding images into DNA
- OligoAnalyzer – *JPEG contribution (JPEG DNA Metrics)* : Software analyzing the quality of the encoded oligos with regard to the biochemical constraints, provided to the JPEG DNA Ad-hoc group as an evaluating software for the JPEG DNA Call for Proposals
- OligoChecker – *JPEG contribution (JPEG DNA Metrics)* : Software checking the validity of the DNA encoded files, provided to the JPEG DNA Ad-hoc group as an evaluating software for the JPEG DNA Call for Proposals
- JPEG DNA SFC4 – *JPEG contribution (Answer to the JPEG DNA Call for Proposals, October 2023)* : JPEG Based image codec adapted to DNA, improvement over the JPEG DNA

Benchmark Codec, submitted to the JPEG DNA Call for Proposals. Several options were available: a source coding (raw input) and a transcoding (binary JPEG input) were provided, with the possibility to add randomization to the encoded data

- HiDNA v1 – *JPEG contribution (Answer to the JPEG DNA Call for Proposals, October 2023)* and v2 : Implicit Neural Representation for Image coding adapted to DNA, the v1 was submitted to the JPEG DNA Call for Proposals
- PIC-DNA – *Contribution to the PEPR Molecularxiv Project by the Data Science Department of EURECOM* : Progressive Image Codec for DNA data Storage based on the J2K progressive codec

Supervision

- 1 Master Semester Project:
 - Enrico Sbuttoni: Master student at EURECOM, Image Progressive Coding for DNA data storage
- 2 co-supervised interns:
 - Aditya Nair: master's trainee from Polytech Nice, co-supervised with Michel Barlaud, Professor at I3S
 - Jérémy Mateos: master's trainee from Université Côte d'Azur, co-supervised with Marc Antonini, Director of research CNRS at I3S

International activities

- Congresses : I have attended 8 congresses during my PhD: EUSIPCO 2022 (Belgrade, Serbia), MWCC 2022 (Munich, Germany), ICASSP 2023 (Rhodes, Greece), DSP 2023 (Rhodes, Greece), GDR ISIS (CNRS) on DNA data storage (Paris, France), JPEG 100th Meeting (Covilha, Portugal), MMSP 2023 (Poitiers, France), ICIP 2023 (Kuala Lumpur, Malaysia)
- I have done 4 stays abroad: a six month research internship in Florence, Italy, in the AILab of Università degli Studi di Firenze and 3 one-week stays in Amsterdam (visit of the Keesing Technologies company), London (Imperial College) and Cambridge, United Kingdom (Linguistic Training)

TEACHING

- Course: DNA-based Image Coding, Image Coding's class of Pr. Jean-Luc Dugelay at EURECOM (2024, 2025)
- Lab Session: RS205 Course, Frederic Payan, Réseaux et Télécommunications department, IUT Université Côte d'Azur (2022)

PUBLICATION LIST

Publications in peer-reviewed journals

- [1] Z. Ziran, X. Pic, S. Undri Innocenti, D. Mugnai, and S. Marinai. "Text alignment in early printed books combining deep learning and dynamic programming". In: *Pattern Recognition Letters* 133 (2020), pp. 109–115.

JPEG public documents

- [2] M. Antonini, M. Dimopoulou, T. Ebrahimi, E. Gil San Antonio, X. Pic, M. Testolina, D. M. Lazzarretto, and A. Pinheiro. "wg1n100418-ICQ-JPEG DNA Exploration Experiment". In: *JPEG 99th Meeting* (2023).
- [3] M. Antonini, T. Ebrahimi, X. Pic, and M. Testolina. "wg1n100597-100-ICQ-Additions to the JPEG DNA Common Test Conditions version 2.0". In: *JPEG 100th Meeting* (2023).

- [4] M. Antonini, L. Cruz, E. da Silva, M. Dimopoulou, S. Ebrahimi Touradj Foessel, E. Gil San Antonio, G. Menegaz, F. Pereira, X. Pic, A. Pinheiro, and M. Raad. "wg1n100098-094-REQ-DNA-based Media Storage: State-of-the-Art, Challenges, Use Cases and Requirements, v7.0". In: *JPEG 94th Meeting* (2022).
- [5] M. Antonini, M. Dimopoulou, X. Pic, and T. Ebrahimi. "wg1n100327-097-ICQ-JPEG DNA Common Test Conditions v1.0". In: *JPEG 97th Meeting* (2022).
- [6] M. Antonini and X. Pic. "wg1n100175-095-ICQ-JPEG DNA Benchmark Codec v0.5". In: *JPEG 95th Meeting* (2022).
- [7] M. Antonini and X. Pic. "wg1n100329-097-ICQ-JPEG DNA Benchmark Codec v1.0". In: *JPEG 97th Meeting* (2022).

Papers in refereed conferences

- [8] D. Lazzarotto, M. Testolina, S. K. Hanna, O. Watanabe, E. G. S. Antonio, X. Pic, M. Dimopoulou, M. Antonini, and T. Ebrahimi. "Overview of JPEG DNA coding system for image storage on synthetic DNA". In: *Applications of Digital Image Processing XLVIII*. Vol. 13605. International Society for Optics and Photonics. SPIE, 2025, 136050P.
- [9] X. Pic and R. Appuswamy. "Combining Progressive Image Compression and Random Access in DNA Data Storage". In: *25th International Conference on Digital Signal Processing (DSP 2025)* (2025).
- [10] X. Pic and R. Appuswamy. "Hierarchical Encoding of JPEG2000-compressed Images for DNA data storage". In: *New Trends in DNA based Data Storage (DBDS 25)* (2025). Poster.
- [11] T. H. Le, X. Pic, J. Mateos, and M. Antonini. "Implicit neural multiple description for DNA-based data storage". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2024)* (2024).
- [12] T. H. Le, X. Pic, and M. Antonini. "INR-MDSQC: Implicit Neural Representation Multiple Description Scalar Quantization for Robust Image Coding". In: *2023 IEEE 25th International Workshop on Multimedia Signal Processing (MMSP)* (2023), pp. 1–6.
- [13] X. Pic, E. G. S. Antonio, M. Dimopoulou, and M. Antonini. "Image storage on synthetic DNA using compressive autoencoders and DNA-adapted entropy coders". In: *2023 IEEE 25th International Workshop on Multimedia Signal Processing (MMSP)* (2023), pp. 1–6.
- [14] X. Pic, E. G. S. Antonio, M. Dimopoulou, and M. Antonini. "Rotating labeling of entropy coders for synthetic DNA data storage". In: *IEEE International Conference on Digital Signal Processing (DSP)* (2023).
- [15] X. Pic, M. Dimopoulou, E. Gil San Antonio, and M. Antonini. "MQ-Coder inspired arithmetic coder for synthetic DNA data storage". In: *30th International Conference on Image Processing (ICIP 2023)* (2023).
- [16] I. Mitrica, E. Gil San Antonio, M. Dimopoulou, X. Pic, and M. Antonini. "Efficient classification of DNA reads for robust decoding of data stored in synthetic DNA Main objective". In: *Munich Workshop on Coding and Cryptography (MWCC 2022)* (2022). Poster.
- [17] X. Pic and M. Antonini. "A constrained Shannon-Fano entropy coder for image storage in synthetic DNA". In: *European Signal Processing Conference (EUSIPCO)* (2022).
- [18] X. Pic and M. Antonini. "Image coding algorithm for DNA data storage combining JPEG and autoencoders". In: *Munich Workshop on Coding and Cryptography (MWCC 2022)* (2022). Poster.

REFERENCES

Available upon request